HOW POPULATIONS EVOLVE

Module map

- Module 13: How Populations Evolve
- Connected lab: lab-14_how-populations-evolve.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 13
How Populations Evolve

1. Allele frequencies as the unit of change
2. Sources of variation
3. Five mechanisms of microevolution
4. Genetic drift and selection
5. Hardy-Weinberg as a baseline

Lab evidence
lab-14_how-populations-evolve.md

HOW POPULATIONS EVOLVE

Learning objectives

- Define microevolution using allele frequencies.
- Compare mutation, gene flow, drift, selection, and the effects of nonrandom mating.
- Distinguish bottleneck and founder effects.
- Interpret directional, stabilizing, and disruptive selection.
- Use Hardy-Weinberg as a non-evolving baseline.

OBJECTIVE 1
Define microevolution using allele frequencies.

OBJECTIVE 2
Compare mutation, gene flow, drift, selection, and the effects of nonrandom mating.

OBJECTIVE 3
Distinguish bottleneck and founder effects.

OBJECTIVE 4
Interpret directional, stabilizing, and disruptive selection.

OBJECTIVE 5
Use Hardy-Weinberg as a non-evolving baseline.

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Topic sequence

- Allele frequencies as the unit of change: Microevolution is allele-frequency change across generations. For diploids, count allele copies and divide by twice the number of individuals.
- Sources of variation: Mutation can create alleles; recombination reshuffles existing alleles. Gene flow moves alleles between reproducing populations.
- Five mechanisms of microevolution: Mutation, flow, drift, and selection can change allele frequencies. Nonrandom mating alone can change genotype proportions without changing alleles.
- Genetic drift and selection: Drift is chance sampling, strongest in small populations. Bottlenecks reduce existing populations; founders start new ones. Selection depends on fitness.
- Hardy-Weinberg as a baseline: Hardy-Weinberg predicts p squared, 2pq, and q squared under explicit assumptions. A mismatch alone neither identifies a cause nor proves allele-frequency change.

01

Allele frequencies as the unit of change

Microevolution is allele-frequency change across generations. For diploids, count allele copies and divide by twice the number of individuals.

02

Sources of variation

Mutation can create alleles; recombination reshuffles existing alleles. Gene flow moves alleles between reproducing populations.

03

Five mechanisms of microevolution

Mutation, flow, drift, and selection can change allele frequencies. Nonrandom mating alone can change genotype proportions without changing alleles.

04

Genetic drift and selection

Drift is chance sampling, strongest in small populations. Bottlenecks reduce existing populations; founders start new ones. Selection depends on fitness.

05

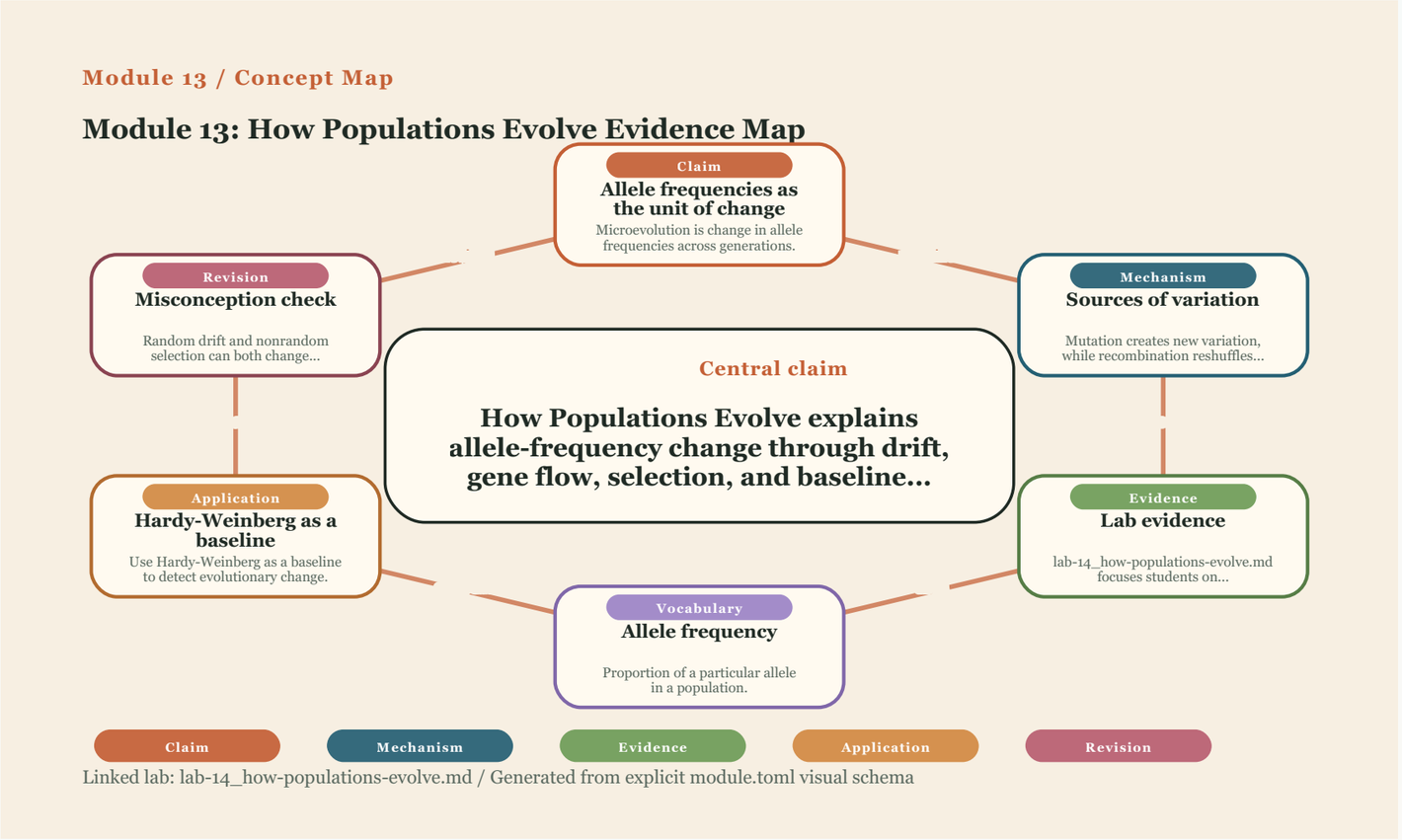
Hardy-Weinberg as a baseline

Hardy-Weinberg predicts p squared, 2pq, and q squared under explicit assumptions. A mismatch alone neither identifies a cause nor proves allele-frequency change.

HOW POPULATIONS EVOLVE

Module 13: How Populations Evolve Evidence Map

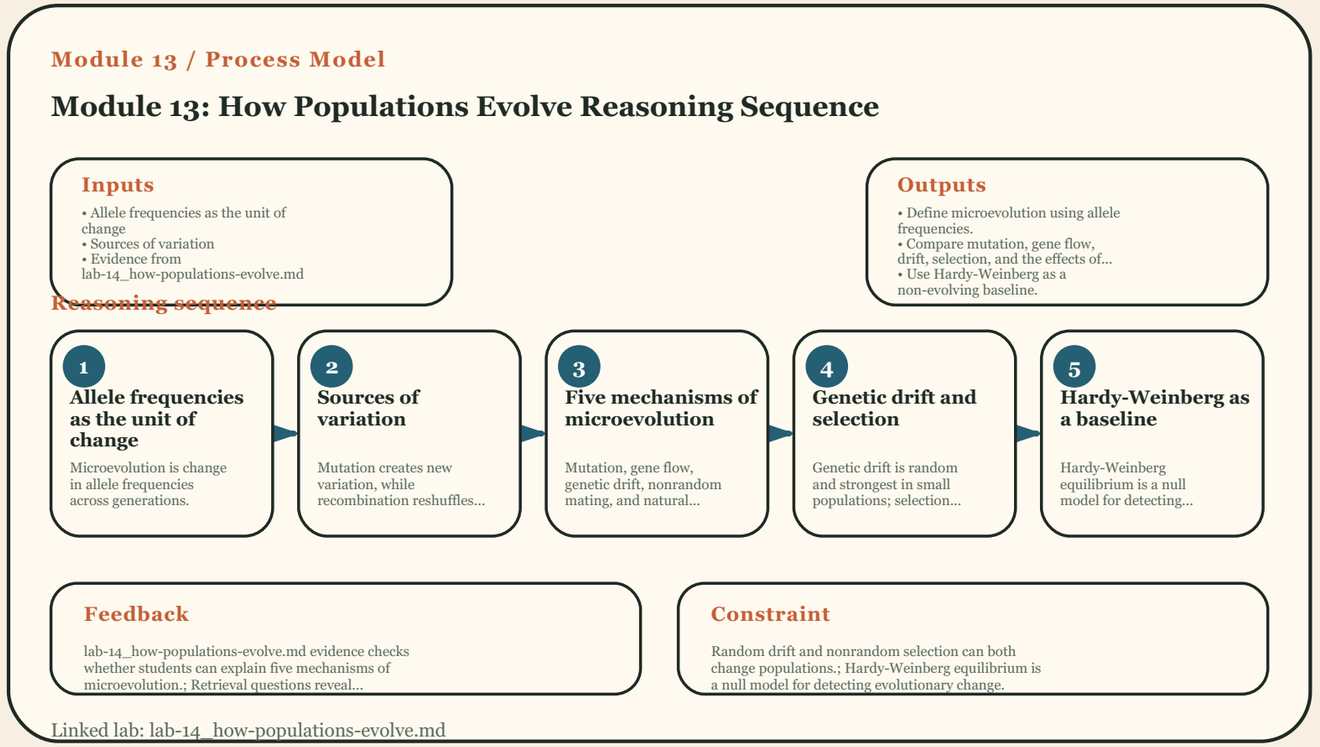
- Module focus: How Populations Evolve
- Central claim: How Populations Evolve explains allele-frequency change through drift, gene flow, selection, and baseline models.
- Key nodes to connect: Allele frequencies as the unit of change, Sources of variation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-14_how-populations-evolve.md supplies an evidence surface for the map.



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Module 13: How Populations Evolve Reasoning Sequence

- Module focus: How Populations Evolve
- Trace the model: Allele frequencies as the unit of change -> Sources of variation -> Five mechanisms of microevolution -> Genetic drift and selection
- Inputs become outputs: Allele frequencies as the unit of change, Sources of variation -> Define microevolution using allele frequencies., Compare mutation, gene flow, drift, selection, and the effects of nonrandom mating.
- Feedback check: lab-14_how-populations-evolve.md evidence checks whether students can explain five mechanisms of microevolution.
- Lab connection: lab-14_how-populations-evolve.md gives students a process to observe or test.



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Terms as evidence handles

- Allele frequency: Proportion of a particular allele in a population.
- Microevolution: Allele-frequency change across generations.
- Mutation: DNA sequence change that can create new alleles.
- Gene flow: Movement of alleles between populations.
- Genetic drift: Random allele-frequency change, strongest in small populations.
- Bottleneck effect: Drift after a sharp population reduction.

ALLELE FREQUENCY
Proportion of a particular allele in a population.

MICROEVOLUTION
Allele-frequency change across generations.

MUTATION
DNA sequence change that can create new alleles.

GENE FLOW
Movement of alleles between populations.

GENETIC DRIFT
Random allele-frequency change, strongest in small populations.

BOTTLENECK EFFECT
Drift after a sharp population reduction.

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Lab connection

- Lab file: lab-14_how-populations-evolve.md
- Lab evidence should test or illustrate: Five mechanisms of microevolution
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Allele frequencies as the unit of change

EVIDENCE
lab-14_how-populations-evolve.md

CLAIM
Define microevolution using allele frequencies.

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Contrast check

- Do not stop at: Microevolution is allele-frequency change across generations. For diploids, count allele copies and divide by twice the number of individuals.
- Stronger explanation: Hardy-Weinberg predicts p squared, 2pq, and q squared under explicit assumptions. A mismatch alone neither identifies a cause nor proves allele-frequency change.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Microevolution is allele-frequency change across generations. For diploids, count allele copies and divide by twice the number of individuals.

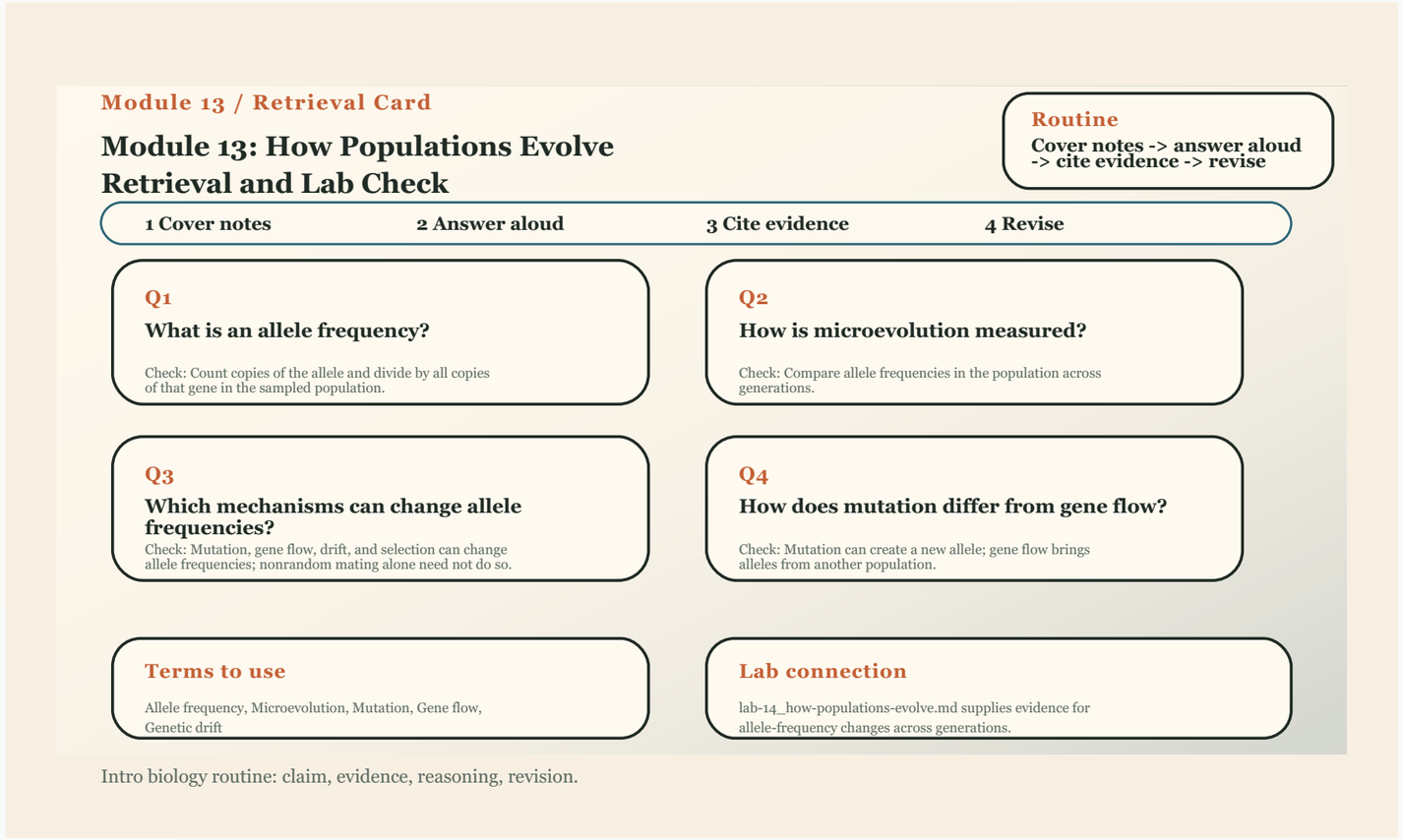
DEEPER

Hardy-Weinberg predicts p squared, 2pq, and q squared under explicit assumptions. A mismatch alone neither identifies a cause nor proves allele-frequency change.

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Module 13: How Populations Evolve Retrieval and Lab Check

- Module focus: How Populations Evolve
- Retrieval prompt: What is an allele frequency?
- Answer check: Count copies of the allele and divide by all copies of that gene in the sampled population.
- Required terms: Allele frequency, Microevolution, Mutation
- Lab connection: lab-14_how-populations-evolve.md; lab-14_how-populations-evolve.md supplies evidence for allele-frequency changes across generations.



Intro biology routine: claim, evidence, reasoning, revision.

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Practice quiz bridge

- 1. Microevolution is defined as:
- 2. Genetic drift has its strongest effect in:
- 3. A few birds colonize a new island and start a population with limited variation. This reduced variation is due to:
- 4. The Hardy-Weinberg principle is useful mainly because it:

QUESTION 1

Answer first

Microevolution is defined as:

QUESTION 2

Answer first

Genetic drift has its strongest effect in:

QUESTION 3

Answer first

A few birds colonize a new island and start a population with limited variation. This reduced variation is due to:

QUESTION 4

Answer first

The Hardy-Weinberg principle is useful mainly because it:

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Synthesis and exit ticket

- Exit claim: explain allele frequencies as the unit of change using evidence from lab-14_how-populations-evolve.md.
- Must include: Allele frequency, Microevolution, Mutation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Allele frequencies as the unit of change

EVIDENCE
lab-14_how-populations-evolve.md

REVISION
What would change your mind?